TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.101	
High Speed Endurance	
Test Carrier:	Roller Drum Test Rig
Testing Facility:	Test Rig of Tire Manufacturer
Status (Porsche):	25.11.2019
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1 Purpose

The purpose of the test procedure is to ensure the high-speed endurance of all tires for the respective vehicle and engine variants according to the valid target data sheet.

2 Field of Application

This test method specifies a procedure for testing the high-speed strength of tires on a chassis dynamometer. The aim is to achieve the test speed specified by Porsche for the respective vehicle and engine variant without air loss.

The test procedure is valid for:

- Test Tire Trials
- Approval Tests
- Series Monitoring

In addition to the Porsche test program, the tire manufacturer must ensure that the tire meets the legal requirements.

3 Definition of Terms

Vmax:
Typed maximum vehicle speed (see target data sheet)

Worst Case Condition High-speed Test:
The most unfavorable load combination of speed, camber, air pressure and axle load selected by the tire manufacturer from the high-speed test program parameter table. The comfort air pressures must also be taken into account.

4 Responsibility and Scope of Testing

Test Tire Trials:
The inspection for the test tire delivery is performed and documented by the tire manufacturer (self-certification). Two tires per specification must be tested consecutively according to the ”worst case” conditions selected by the tire manufacturer. The results are made available to the PAG immediately.

Approval Tests:
The test for approval is performed and documented by the tire manufacturer (self-certification). For each specification to be released, two tires must be tested in succession according to the ”worst case” conditions selected by the tire manufacturer. The results are made available to the PAG immediately.

Series Monitoring:
The monitoring of the series is carried out and documented by the tire manufacturer (self-certification). The test intervals and the number of test tires are at the discretion of the tire manufacturer. However, at least once a year two tires should be tested consecutively according to the ”worst case” conditions selected by the tire manufacturer and the result should be reported to the PAG. In case of non-compliance, the PAG must be informed immediately.

Retesting:
Retesting of release/production tires can be arranged by the PAG, Tire Department, at an independent institute.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

Measured Variable	**Measurement Tolerance**	**Adjustment & Control Tolerance**
Drum Speed	±1% of measurand	±1 km/h
Wheel Load	±1% of measurand	±50 N
Tire Pressure	±0.1 bar	±0.1 bar
Ambient Temperature	±0.5° C	-
Tire Surface Temp.	±0.5° C	-
Time	±1 s	-

The tire pressure is measured at the beginning of the measurement. The tire surface temperature, if not continuously measurable, is measured at the beginning and if required (e.g. for the release test) at the end or up to three minutes after the end of the measurement. The ambient temperature is measured continuously.

5.1.2 Test Rig

- External Drum Test Rig with diameter 1.7m, 2.0m or 3.0m
- Datum diameter with 2.0m (different load reductions depending on diameter, see test parameters)
- Drum surface: steel, smooth
- Drum driven, speed controlled
- Wheel receiver with camber adjustment possibility up to ±4°

5.1.3 Data Processing

N/A

5.2 Test Conditions

- Ambient temperature must be between 20°C and 30°C
- Reference temperature for air pressure setting is 20°C
- The tire manufacturer has to ensure all vehicle and air pressure variants by his tests on the basis of the valid target data sheet (parameter table for the high-speed test program)

5.3 Test Carrier / Test Set-Up

- The test load corresponds to the wheel load selected according to the worst-case condition in the target data sheet (parameter table for the high-speed test program) multiplied by:
- 0.9, if drum diameter = 1.7m or 2.0m
- 0.93, if drum diameter = 3.0m
- Tire pressure unregulated (see target data sheet)
- Camber angle depending on the respective vehicle value (see target data sheet)
- Skew angle = 0
- Rim width, diameter and contour at the rim seat of the test rim must correspond to the respective original Porsche wheel (see target data sheet)
- Additional drillings e.g. for internal temperature measurements are permissible

5.4 Test Procedure

The mounted wheel is stored under test room temperature for at least three hours before testing. Before the test, the tire pressure is adjusted with the wheel loaded. The drum speed is increased gradually according to the respective program in the following table. The acceleration phases of the drum should not exceed 30 seconds (one minute at the beginning).

Test Stages	**Runtime (min)**	**Drum speed (km/h)**
1	5	Vmax - 90
2	5	Vmax - 40
3	10	Vmax - 30
4	10	Vmax - 20
5	10	Vmax - 10
6	20	Vmax
7	20	Vmax + 10
8	20	Vmax + 20

End of the test: Basically, all speed levels must be passed. For details see under 6.3 Requirements.

6 Documentation

6.1 Data Analysis

The following data must be listed in the results sheet:

- Test date and time
- Tire size
- Tire make, designation and specification number
- DOT number
- Tire condition (new, used, tread depth)
- Designation of the test program
- Designation of the selected vehicle, loading and air pressure scenario, or alternatively a selected combination of wheel load, camber air pressure and speed as a ”worst case” condition
- Documentation of the target data sheet used for testing (date/change status)
- Test parameters (load, air pressure, wheel dimensions, camber)
- Speed level at 1st damage occurrence
- Runtime at 1st damage occurrence
- Speed level at test termination
- Runtime at test termination
- Cause of damage, damage pattern (attach at least on photo)
- Ambient temperature per speed level and at the end of the test
- Tire surface temperature at the beginning and at the end of the test
- Tire pressure at the beginning and at the end of the test
- Name of the tester
- Test site, test rig

6.2 Presentation of the results

The presentation of the results for test tires as well as for the release test 2U or BMG release is to be executed in the form of the test catalog, data sheet high-speed test F.360.101 for tire releases (release tests) of the PAG.

6.3 Requirements

Requirements Test Tires
The high-speed test for the delivery is considered as passed if test stages 1 to 7 (equivalent to 80 minutes runtime) are passed without damage. In test stage 8, the first tire damage may occur as long it does not lead to sudden air loss (e.g. blisters, cracks, block shedding, tread separation, or compound burnout in the tread block). The defect pattern that has occurred must be documented. To protect the test equipment, the test in test stage 8 may be discontinued after the first tire damage has occurred.

Requirements Approval Tests
The high-speed test for the tire approval (approval tests) is considered as passed if test stages 1 to 7 (equivalent to 80 minutes runtime) are passed without damage. In test stage 8, the first tire damage may occur as long it does not lead to sudden air loss (e.g. blisters, cracks, block shedding, tread separation, or compound burnout in the tread block), however, the test may not be canceled. The runtime of 20 minutes in test level 8 must be met. During the remaining time after the occurrence of a first tire damage, there must be no sudden loss of air. The defect pattern that has occurred must be documented. After the approval test has been carried out, a tire certificate must be sent to PAG without being requested to do so.

Requirements Series Monitoring
The high-speed test for the series monitoring (not approval tests) is considered as passed if test stages 1 to 7 (equivalent to 80 minutes runtime) are passed without damage. In test stage 8, the first tire damage may occur as long it does not lead to sudden air loss (e.g. blisters, cracks, block shedding, tread separation, or compound burnout in the tread block). The defect pattern that has occurred must be documented. To protect the test equipment, the test in test stage 8 may be discontinued after the first tire damage has occurred.

7 Applicable Documents

Porsche AG, data sheet: ”High-Speed Test - F.360.101” Tire Certificate

8 Distributor

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